Pauli's Heuristic as Statistical Inference: The Exclusion Principle as a Parsimonious Four-Parameter Model, 1913–1928

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Abstract

In his 1925 paper on the "complex structure of spectra," Wolfgang Pauli proposed that each electron in an atom is characterised by exactly four quantum numbers—the principal quantum number n, the azimuthal (orbital) quantum number l, the magnetic quantum number m_l, and a newly introduced fourth, "two-valued" quantum number m_s—and that no two electrons in any atom may share the same 4-tuple (n, l, m_l, m_s). This note examines Pauli's reasoning from a modern statistical perspective. The empirical regularities that Pauli organised—electron shell capacities of 2, 8, 18, 32; the doublet structure of alkali spectra; the anomalous Zeeman effect; the X-ray spectroscopic data systematised by Stoner (1924)—constitute a discrete dataset of counts and multiplicities. Pauli's four-parameter state space predicts those counts *exactly* via the formula: capacity of subshell l equals $2(2l+1)$, implying a total shell capacity of $2n^2$. We frame this as a highly constrained statistical model in which the parameters are the allowed ranges of four discrete quantum numbers and a binary occupation constraint, and we examine which "data" identified each parameter. Pauli's procedure parallels Einstein's photoelectric derivation: theoretical insight restricted the hypothesis class to a parsimonious form, reducing what might have been an unmanageable combinatorial estimation problem to a counting exercise whose predictions match every relevant observation. The episode illustrates how a one-sentence theoretical postulate—the Exclusion Principle—functions as an extreme form of inductive bias, compressing the effective hypothesis space from an open-ended combinatorial

family to a single, fully determined prediction with zero tunable parameters.

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1 Introduction

In December 1900, Max Planck introduced the energy quantum $h\nu$ to fit the blackbody spectrum. Five years later, Einstein applied the same quantum to the photoelectric effect, deriving a two-parameter linear equation $V_0 = (h/e)\nu - W/e$ whose slope encoded Planck's constant. Twenty years after that, in January 1925, Wolfgang Pauli proposed something structurally similar: a minimal theoretical principle from which a large body of empirical facts about atomic spectra and the periodic table follows as a counting exercise.

Pauli's principle—that no two electrons in an atom may occupy the same quantum state—is presented in textbooks as a law of nature, delivered with an air of inevitability. The historical record tells a more interesting story. Pauli *conjectured* the principle as the simplest model consistent with a specific collection of quantitative empirical regularities that had accumulated between 1913 and 1924. He called his fourth quantum number a "classically non-describable two-valuedness" [*eine nicht klassisch beschreibbare Zweideutigkeit*] because he had no physical interpretation for it. It was, in his own words, a formal device introduced "at any cost" to organise the data. In this respect, Pauli's reasoning is a closer relative of Planck's 1900 "act of desperation" than of the theoretical deductions from first principles that characterise later quantum mechanics.

This note examines Pauli's 1925 paper through the lens of modern statistical inference. The central questions are:

1. What were the empirical *data* that Pauli was organising?
2. What is the four-parameter *model* that his principle defines, and how many distinct states does it predict for each shell and subshell?
3. Which parameter—or which constraint—is identified by which observation?
4. How does the episode compare, as an exercise in inference and hypothesis formation, with Einstein's derivation of the photoelectric equation?

Sections 2–3 sketch the experimental history from Bohr's 1913 model to Stoner's 1924 consolidation. Section 4 defines Pauli's four-parameter state space precisely and works through the counting algebra. Section 5 assembles the empirical "dataset"—the shell capacities, spectral multiplicities, and anomalous Zeeman observations—that the model is required to fit. Section 6 examines the goodness of fit and the identification of each parameter from distinct empirical observations. Section 7 discusses the identification logic: which data identify each parameter. Section 8 frames the counting problem as a maximum likelihood estimation problem over a discrete state space. Section 9 traces the subsequent confirmations (Uhlenbeck and Goudsmit's spin hypothesis; Fermi–Dirac statistics; Dirac's relativistic derivation; the spin–statistics theorem). Section 10 reinterprets the episode in the language of machine learning. Section 11 compares Pauli's episode with Einstein's photoelectric derivation. Section 13 concludes.

2 Historical Context, 1913–1924

2.1 The Bohr Model and Its Three Quantum Numbers

Niels Bohr published his model of the hydrogen atom in 1913 [Bohr, 1913]. The key postulate was that the electron's orbital angular momentum is quantised: $L = n\hbar$, $n = 1, 2, 3, \ldots$ This single integer, the *principal quantum number* n, gives the energy levels

$$E_n = -\frac{13.6\,\mathrm{eV}}{n^2}, \tag{1}$$

correctly predicting the hydrogen emission spectrum (Lyman, Balmer, and Paschen series) to parts per thousand.

Arnold Sommerfeld extended Bohr's model in 1916 [Sommerfeld, 1916] by allowing elliptical orbits and applying the Wilson–Sommerfeld quantisation condition $\oint p_i\, dq_i =$

$n_i h$ to each coordinate. This introduced two additional integers:

- The *azimuthal quantum number* l (Sommerfeld's k, relabelled later): $l \in \{0, 1, \ldots, n-1\}$, determining the orbital angular momentum magnitude $|\mathbf{L}| = \sqrt{l(l+1)}\,\hbar$.
- The *magnetic quantum number* m_l: $m_l \in \{-l, -l+1, \ldots, +l\}$, giving the projection $L_z = m_l \hbar$ of orbital angular momentum on an external magnetic field axis. In a field, each m_l value produces a slightly different energy (Zeeman splitting).

With these three integers (n, l, m_l) the number of distinct states in shell n is

$$\sum_{l=0}^{n-1} (2l+1) = n^2. \tag{2}$$

Table 1 lists these counts for the first four shells.

Table 1: States per shell under the three-quantum-number Bohr–Sommerfeld model vs. observed electron capacities. The observed capacity equals $2n^2$—exactly twice the predicted n^2—a discrepancy that motivated Pauli's fourth quantum number.

Shell	n	Predicted states (n^2)	Observed capacity ($2n^2$)
K	1	1	2
L	2	4	8
M	3	9	18
N	4	16	32

The predicted capacities fall short by a factor of exactly 2. This was known but unexplained before 1925.

2.2 Failures of the Three-Parameter Model

Despite its successes for hydrogen, the Bohr–Sommerfeld three-quantum-number scheme faced three acute empirical failures by 1924.

Shell closures and the periodic table. Bohr proposed in 1922 that the chemical periodic table is organised by fills of successive electron shells, with noble gases corresponding to completed shells: helium has 2 electrons (shell $n = 1$ full), neon has 10 (shells $n = 1, 2$ full), argon has 18 ($n = 1, 2, 3$ partially full in the $3s, 3p$ subshells),

krypton has 36, and xenon has 54. No dynamical principle within the three-quantum-number scheme explained why a shell can hold exactly $2n^2$ electrons—rather than n^2, or $3n^2$, or any other number—or why it is energetically "closed" once filled.

The anomalous Zeeman effect. Pieter Zeeman discovered in 1896–97 [Zeeman, 1897] that magnetic fields split spectral lines. Classical electron theory (Lorentz) predicted the *normal Zeeman effect*: each line splits into exactly three components ($\Delta m_l = 0, \pm 1$). In reality, most lines in multi-electron atoms split into *more* than three components, with fractional-unit spacings. This *anomalous Zeeman effect* was inexplicable within the three-quantum-number scheme. Pauli in 1924 wrote the most famous frustrated lament in the history of atomic physics, reportedly saying:

> "How can one look happy when he is thinking about the anomalous Zeeman effect?" [Pais, 1991]

Half-integer angular momenta. Alfred Landé in 1921 [Landé, 1921] fitted the anomalous Zeeman patterns empirically with the formula

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}, \tag{3}$$

where J is the total angular momentum quantum number and L, S denote orbital and "spin" contributions. This formula reproduced the observed line-splitting ratios remarkably well, but required J to take *half-integer* values for many atomic states—impossible in the three-quantum-number scheme, where all angular momenta are integers.

3 Stoner's 1924 Consolidation

The immediate empirical precursor to Pauli's paper was a 1924 article by Edmund Clifton Stoner [Stoner, 1924]. Stoner used X-ray spectroscopic level data (from Siegbahn and collaborators) to count the number of electrons assigned to each sublevel of each shell.

Stoner's key observation: in an external magnetic field, all magnetic sublevels (m_l values) within a given (n, l) subshell become energetically distinct. X-ray experiments

showed that the number of electrons per subshell is

$$\text{capacity of subshell } (n, l) = 2(2l + 1). \tag{4}$$

The factor $2l + 1$ counts the distinct m_l values; each m_l sublevel holds *two* electrons. The mysterious factor of 2 per m_l sublevel could not be explained by any then-known quantum number.

Summing equation (4) over $l = 0, 1, \ldots, n - 1$ gives

$$\text{capacity of shell } n = \sum_{l=0}^{n-1} 2(2l + 1) = 2 \sum_{l=0}^{n-1} (2l + 1) = 2n^2. \tag{5}$$

This matches the observed noble-gas configurations exactly (Table 1). But Stoner did not explain *why* each m_l sublevel holds two electrons. Pauli read Stoner's paper and recognised the answer: each sublevel is doubly occupied because there exists a fourth, two-valued quantum number that takes two distinct values. Occupancy is then one electron per distinct 4-tuple, and the factor of 2 follows.

4 Pauli's Four-Parameter Model

4.1 The Four Quantum Numbers

Pauli's 1925 paper [Pauli, 1925] introduces the following model. Each electron in an atom is completely characterised by four quantum numbers:

1. **Principal quantum number** $n \in \{1, 2, 3, \ldots\}$: determines the shell and (for hydrogen) the energy $E_n \propto -1/n^2$. Range: any positive integer.

2. **Azimuthal quantum number** $l \in \{0, 1, \ldots, n - 1\}$: determines the subshell and the orbital angular momentum magnitude. Range: n possible values for each n.

3. **Magnetic quantum number** $m_l \in \{-l, -l+1, \ldots, +l\}$: projection of orbital angular momentum on the field axis; determines the normal Zeeman sublevel. Range: $2l + 1$ possible values for each l.

4. **Fourth quantum number** $m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$: a new, purely quantum-mechanical,

two-valued property of the electron with no classical mechanical analogue. Pauli called it a *Zweideutigkeit*—two-valuedness. Range: exactly 2 values.

Pauli could not give a physical interpretation for m_s in 1925; he was explicit that it is "not classically describable." The physical interpretation (intrinsic electron spin) was supplied a few months later by Uhlenbeck and Goudsmit [1925].

4.2 The Exclusion Constraint

With the four quantum numbers defined, the Exclusion Principle states:

> **Pauli's Exclusion Principle (1925).** In a multi-electron atom, no two electrons may simultaneously have identical values of all four quantum numbers (n, l, m_l, m_s).

Equivalently, each distinct 4-tuple (n, l, m_l, m_s) is a quantum *state*, and every state may be occupied by at most one electron. The occupation number of state s satisfies

$$n_s \in \{0,\, 1\}. \tag{6}$$

This binary constraint is the statistical mechanics analogue of a $\{0, 1\}$ dummy variable. It is the defining property of *fermions*—particles that obey Fermi–Dirac statistics—in contrast to *bosons*, for which occupation numbers are unrestricted non-negative integers.

4.3 Counting States: The Model's Predictions

States per subshell. For fixed (n, l), the number of distinct states is

$$\underbrace{(2l+1)}_{\text{values of } m_l} \times \underbrace{2}_{\text{values of } m_s} = 2(2l+1). \tag{7}$$

Under the Exclusion Principle, the maximum number of electrons in the (n, l) subshell equals the number of distinct states:

$$\text{capacity}(n, l) = 2(2l+1). \tag{8}$$

Table 2 lists the first four subshell types.

Table 2: Predicted subshell capacities from Pauli's four-parameter model, equation (8). Each entry agrees exactly with observation.

Subshell	l	m_l values	States $2(2l+1)$	Observed electrons
s	0	$\{0\}$	2	2
p	1	$\{-1, 0, +1\}$	6	6
d	2	$\{-2, -1, 0, +1, +2\}$	10	10
f	3	$\{-3, -2, -1, 0, +1, +2, +3\}$	14	14

States per shell. For fixed n, summing over subshells:

$$\text{capacity}(n) = \sum_{l=0}^{n-1} 2(2l+1) = 2\sum_{l=0}^{n-1}(2l+1) = 2n^2. \tag{9}$$

Derivation: $\displaystyle\sum_{l=0}^{n-1}(2l+1) = 2 \cdot \frac{(n-1)n}{2} + n = n^2$, so the shell capacity is $2n^2$.

Table 3 verifies these predictions against noble-gas electron configurations. The agreement is exact at every shell.

Table 3: Shell capacities predicted by Pauli's model vs. observed electron configurations of noble gases. "Observed" electron count is the total for the filled inner shells, confirmed by X-ray spectroscopy and the chemical inertness of the noble gases.

Shell	n	Predicted $2n^2$	Observed	Noble gas
K	1	2	2	He ($Z = 2$): $1s^2$
L	2	8	8	Ne ($Z = 10$): He + $2s^2 2p^6$
M	3	18	18	Ar ($Z = 18$): Ne + $3s^2 3p^6$
N	4	32	32	Kr ($Z = 36$): Ar + $4s^2 3d^{10} 4p^6$

5 The Empirical Dataset

The "data" that Pauli was required to organise fall into three classes, each contributing a distinct type of *empirical count* or *empirical multiplicity*.

5.1 Class 1: Shell and Subshell Electron Capacities

The electron capacities of each shell and subshell were known empirically from two sources:

1. *Chemical valence and noble-gas stability.* Bohr in 1922 inferred from the periodic table that the noble-gas configurations correspond to closed shells with 2, 8, 18, 32 electrons.

2. *X-ray spectroscopy.* Manne Siegbahn's group measured the binding energies and number of energy sublevels in the K, L, M, and N shells, identifying how many electrons reside at each distinct energy. Stoner [1924] systematised these in his 1924 paper.

The resulting empirical dataset is given in Table 4. Each row is one "observation."

Table 4: Empirical electron capacities used by Pauli. Shell and subshell electron counts from chemical data (noble-gas configurations) and X-ray spectroscopy (Siegbahn; systematised by Stoner 1924). These are the observables that Pauli's model must predict exactly.

Shell	n	Subshell	l	Observed electrons
K	1	$1s$	0	2
L	2	$2s$	0	2
		$2p$	1	6
M	3	$3s$	0	2
		$3p$	1	6
		$3d$	2	10
N	4	$4s$	0	2
		$4p$	1	6
		$4d$	2	10
		$4f$	3	14

5.2 Class 2: Spectral Term Multiplicities

The *multiplicity* of a spectral term $^{2S+1}L_J$ counts the number of distinct quantum states with a given L, S, and J: specifically, it equals $2J + 1$ (the number of m_J projections). The observed term multiplicities for the first few elements are:

- *Alkali metals* (Li, Na, K, ...): doublet spectra ($^2S_{1/2}$, $^2P_{1/2}$, $^2P_{3/2}$, ...). The doublet character—two lines everywhere the classical single-electron theory predicted

one—was a direct signal of $S = \frac{1}{2}$.

- *Alkaline earths* (Be, Mg, Ca, ...): singlet and triplet series ($^1S_0$, $^3S_1$, $^1P_1$, $^3P_{0,1,2}$, ...).
- *Carbon, nitrogen, oxygen*: quartet terms among excited configurations.

These term multiplicities are predicted exactly by the four-quantum-number scheme: for a closed subshell the total spin is zero (singlet); for an outer shell with one electron the total spin is $S = \frac{1}{2}$ (doublet); for two valence electrons in different orbitals, $S \in \{0, 1\}$ (singlet or triplet).

5.3 Class 3: Anomalous Zeeman Component Counts

In a weak external magnetic field, each spectral term $^{2S+1}L_J$ splits into $2J+1$ sublevels. A transition $^{2S+1}L_J \to {}^{2S'+1}L'_{J'}$ then produces a number of distinct emission lines equal to the number of allowed ($|\Delta m_J| \leq 1$, $\Delta m_J \in \{-1, 0, +1\}$) transitions between those sublevels. Table 5 lists key observed anomalous Zeeman patterns.

Table 5: Observed anomalous Zeeman component counts for selected sodium spectral lines. The upper and lower J values and the corresponding m_J-sublevel counts are listed. "Predicted" uses the four-quantum-number model with the exclusion principle and is identical to the observed count in every case. Classical Lorentz theory always predicts 3 components.

Transition	Line (nm)	Upper J	Lower J	Observed lines	Classical
$^2P_{3/2} \to {}^2S_{1/2}$	589.0 (D_2)	3/2	1/2	6	3
$^2P_{1/2} \to {}^2S_{1/2}$	589.6 (D_1)	1/2	1/2	4	3

Calculation for the D_2 line. The upper state $^2P_{3/2}$ has $J = 3/2$, giving $m_J \in \{-3/2, -1/2, +1/2, +3/2\}$ (4 sublevels). The lower state $^2S_{1/2}$ has $J = 1/2$, giving $m_J \in \{-1/2, +1/2\}$ (2 sublevels). The allowed transitions with $\Delta m_J \in \{-1, 0, +1\}$ number exactly 6, as observed. For the D_1 line, both levels have $J = 1/2$; the 4 allowed transitions are ($\Delta m_J = 0, \pm 1$) from 2 upper to 2 lower sublevels (one $\Delta m_J = 0$ is forbidden for $J = 1/2 \to 1/2$, $\Delta J = 0$ by the selection rule, leaving 4). [1]

[1] Precisely: for $\Delta J = 0$ the transition $m_J = 0 \to m_J = 0$ is allowed only if $J \neq 0$. For the D_1 line the two $\Delta m_J = 0$ transitions both occur ($+1/2 \to +1/2$ and $-1/2 \to -1/2$), giving 2 longitudinal lines, combined with 1 transverse σ^+ and 1 σ^-, totalling 4.

These counts—6 and 4 instead of the classical 3—are *impossible* within the three-quantum-number scheme, which yields only integer J and hence cannot produce the half-integer $J = 3/2$ or $J = 1/2$ occurring in the sodium D doublet.

5.4 Landé's g-Factor Data

Landé's 1921 empirical formula (3) summarises a further dataset: the *spacing ratios* of Zeeman components. The splitting of a level in a magnetic field B is $\Delta E = g_J \mu_B m_J B$, where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton. Observed g_J values for sodium are:

$$\begin{aligned} g_J(^2S_{1/2}) &= 2.0, \\ g_J(^2P_{1/2}) &= 2/3 \approx 0.667, \\ g_J(^2P_{3/2}) &= 4/3 \approx 1.333. \end{aligned}$$

These non-integer values—inexplicable if $g_{\text{spin}} = 1$ as classical orbital mechanics would have it—require $g_s = 2$ for the spin contribution. Formula (3) with $g_s = 2$ reproduces all observed g_J values exactly. The anomalous value $g_s = 2$ is identified by the Zeeman spacing data.

6 Goodness of Fit: Zero Residuals

Table 6 compares Pauli's predicted electron capacities with every observed capacity from Table 4.

The residual sum of squares is identically zero. A formal χ^2 goodness-of-fit test statistic

$$\chi^2 = \sum_{\text{subshells}} \frac{(\text{Observed} - \text{Predicted})^2}{\text{Predicted}} = 0.$$

Pauli's model does not fit the data approximately; it generates the observed counts *deductively*. The fitting exercise has zero degrees of freedom in the residual because the model produces the observed counts as exact consequences of counting arguments, not as the output of a curve-fit with estimated free parameters.

This situation precisely parallels Einstein's photoelectric equation: the five-point OLS regression for sodium gave $R^2 \approx 0.99999$ and residuals below 2 mV, but crucially the

Table 6: Predicted vs. observed subshell electron capacities. Pauli's model predicts $2(2l+1)$ electrons per subshell (n, l). The residual is zero in every case; the model fits all ten observations exactly with no adjustable parameters beyond the ranges of the four quantum numbers.

Subshell	l	Formula $2(2l+1)$	Predicted	Observed	Residual
$1s$	0	2(1)	2	2	0
$2s$	0	2(1)	2	2	0
$2p$	1	2(3)	6	6	0
$3s$	0	2(1)	2	2	0
$3p$	1	2(3)	6	6	0
$3d$	2	2(5)	10	10	0
$4s$	0	2(1)	2	2	0
$4p$	1	2(3)	6	6	0
$4d$	2	2(5)	10	10	0
$4f$	3	2(7)	14	14	0
RSS					0

functional form $V_0 = \alpha + \beta\nu$ and the *universality of the slope* $\beta = h/e$ were *predicted* by theory, not estimated from data. For Pauli, the functional form of the capacity formula $2(2l+1)$ is similarly predicted by theory; the counting data serve as a confirmation test, not as inputs to a parameter-estimation algorithm.

7 Parameter Identification

In a classical regression, "identification" asks which variation in the data pins down which parameter. For Pauli's lattice of allowed states, the analogous question is: which empirical observation establishes each of the four quantum numbers and the exclusion constraint?

Parameter	What it determines	Identifying observation
$n \in \{1, 2, 3, \ldots\}$	Shell index; number of subshells $= n$	Chemical periodicity; discrete X-ray absorption edges
$l \in \{0, \ldots, n-1\}$	Subshell type; $2l+1$ magnetic sublevels	Fine structure of X-ray levels; Sommerfeld's elliptical orbits
$m_l \in \{-l, \ldots, +l\}$	Normal Zeeman splitting into $2l+1$ components	Normal Zeeman effect (integer-valued splitting, $\Delta\nu \propto m_l$)
$m_s \in \{\pm\frac{1}{2}\}$	Factor-of-2 per m_l sublevel; doublet structure in alkali spectra	Stoner's factor-of-2 per X-ray sublevel; alkali-metal doublets; Landé $g_s = 2$
Exclusion ($n_s \leq 1$)	Total shell capacity $= 2n^2$; chemical inertness of noble gases	Noble-gas stability; absence of transitions out of closed shells

Remark on the exclusion constraint. Without the constraint $n_s \leq 1$, multiple electrons could pile into the ground state $1s$ (as bosons do). Every atom would have all its electrons in the $n = 1$, $l = 0$, $m_l = 0$, $m_s = +\frac{1}{2}$ state, and the periodic table would not exist. The chemical diversity of matter is a consequence solely of this constraint. In statistical language, the constraint is identified by the existence of the periodic table itself: the chemical fact that atoms do not all collapse to a single ground-state configuration is direct evidence that electrons must obey an exclusion rule.

Remark on m_s. The identification of m_s is notably *indirect*: the empirical evidence for a fourth quantum number comes from its *consequences* (the factor of 2 in subshell capacities; the doublet line structure) rather than from a direct measurement of the quantum number itself as a physical observable. Pauli acknowledged this explicitly in his Nobel lecture: "I was unable to give a logical reason for the exclusion principle or to derive it from more general assumptions" [Pauli, 1964]. The physical object corresponding to m_s was identified only after Uhlenbeck and Goudsmit proposed electron spin.

8 A Maximum Likelihood Framing

The photoelectric regression has a natural MLE formulation under Gaussian errors. Pauli's counting problem is discrete, and its counterpart is a multinomial likelihood. We set this up here to make the statistical parallel explicit.

8.1 The Discrete Probability Model

Consider the experiment of drawing an electron uniformly at random from among all quantum states available in shell n, and recording its (l, m_l, m_s) values. Under the Exclusion Principle, each available state is equally accessible (the Pauli principle says nothing about energy ordering; for the purpose of counting, all states are equivalent). The relevant statistical object is therefore the *marginal probability* that the electron is in subshell l:

$$p(l \mid n) = \frac{2(2l+1)}{2n^2} = \frac{2l+1}{n^2}, \quad l = 0, 1, \ldots, n-1. \tag{10}$$

This is a discrete distribution on $\{0, 1, \ldots, n-1\}$ with probabilities proportional to the subshell degeneracy $2l + 1$.

8.2 The Multinomial Likelihood

Suppose we observe N_l electrons in subshell l for $l = 0, 1, \ldots, n-1$, with $\sum_l N_l = N$ (the total number of electrons in shell n at maximum fill). The multinomial likelihood is

$$\mathcal{L}(\{p_l\}) \propto \prod_{l=0}^{n-1} p_l^{N_l}. \tag{11}$$

Pauli's model predicts the *unique* probability vector $p_l^* = (2l+1)/n^2$. Any competing model that, for example, assigns equal probability to all subshells ($p_l = 1/n$) or uses only three quantum numbers ($p_l = 1/(n^2)$ per m_l if no m_s) would predict different capacities.

Under the true multinomial, the log-likelihood ratio comparing Pauli's model to the null model of equal subshell occupancy ($p_l^{\text{null}} = 1/n$ for all l) is

$$\Lambda = 2\sum_{l=0}^{n-1} N_l \ln\frac{p_l^*}{p_l^{\text{null}}} = 2N\sum_{l=0}^{n-1} \frac{2l+1}{n^2} \ln\frac{n(2l+1)}{n^2}, \tag{12}$$

which is strictly positive (Pauli's model fits better) for all $n \geq 2$. The χ^2 statistic comparing the predicted capacities from Table 6 with the null prediction of $2n^2/n$ electrons per subshell is:

Shell n	Subshell l	Predicted $(2(2l+1))$	Null $(2n^2/n = 2n)$	χ^2 contribution
$n = 2$	$l = 0$ (s)	2	4	$4/4 = 1.00$
$n = 2$	$l = 1$ (p)	6	4	$4/4 = 1.00$
			Shell $n = 2$ total χ^2	2.00
$n = 3$	$l = 0$ (s)	2	6	$16/6 = 2.67$
$n = 3$	$l = 1$ (p)	6	6	$0/6 = 0.00$
$n = 3$	$l = 2$ (d)	10	6	$16/6 = 2.67$
			Shell $n = 3$ total χ^2	5.33

The null model of equal subshell occupancy is clearly rejected by the observed capacity data at any conventional significance level. Pauli's model, by contrast, fits perfectly.

8.3 Parsimony: The Model Has No Free Parameters

The multinomial likelihood for the capacity data has, in principle, $n - 1$ free probability parameters for each shell n. Pauli's model reduces these to *zero* free parameters: the probabilities $p_l^* = (2l + 1)/n^2$ are completely determined by the structure of the four quantum numbers and the counting rule. There is nothing to estimate.

In the language of information criteria, if a competitor model uses k free parameters and Pauli's model uses 0, the Akaike Information Criterion (AIC) favours Pauli's model by $2k$ units *plus* the improvement in log-likelihood from the exact fit. Since the exact fit achieves the maximum possible likelihood (the observed data are the maximum-likelihood estimate of their own probabilities), no model can outpredict Pauli's on data that were used to derive the principle — but Pauli's model generalises analytically to any shell, including $n = 5$ (xenon) and $n = 7$ (actinides), whereas a purely empirical fit to shells $n = 1, \ldots, 4$ would have no basis for predicting shell 5.

9 Subsequent Confirmations

9.1 Uhlenbeck and Goudsmit: Physical Interpretation of m_s

Pauli proposed m_s as a purely formal device in January 1925. In September 1925, George Uhlenbeck and Samuel Goudsmit [1925] proposed that the electron possesses an *intrinsic angular momentum* (spin) of magnitude $\hbar/2$, with projection $m_s = \pm\frac{1}{2}$ along any axis. The associated magnetic moment has an anomalous gyromagnetic ratio $g_s \approx 2$—twice the classical orbital value. The Uhlenbeck–Goudsmit proposal:

- Gave a physical object—intrinsic spin—to Pauli's abstract two-valuedness.
- Explained the doublet structure of alkali spectra as arising from the spin–orbit coupling $\mathbf{L} \cdot \mathbf{S}$ between orbital and spin angular momenta.
- Reproduced the Landé g-factor formula (3) with the substitution $g_s = 2$, explaining why the formula contains the factor 2 in the numerator.

9.2 Fermi–Dirac Statistics

In 1926 Enrico Fermi [1926] and Paul Dirac [1926] independently derived the statistical mechanics of a gas of identical particles obeying the Exclusion Principle. For a system in thermal equilibrium at temperature T with chemical potential μ, the mean occupation number of a state with energy ε is

$$\bar{n}(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/k_B T} + 1}. \tag{13}$$

The $+1$ in the denominator (vs. the -1 of Bose–Einstein statistics) is the signature of the Exclusion Principle: the constraint $n_s \leq 1$ prevents accumulation, and $0 < \bar{n} < 1$ for all ε and $T > 0$. At $T = 0$, equation (13) becomes a step function: all states below the Fermi energy $E_F = \mu(T = 0)$ are completely filled ($\bar{n} = 1$) and all above are empty ($\bar{n} = 0$)—a direct restatement of the Exclusion Principle as a thermodynamic limiting case.

Fermi–Dirac statistics had immediate astrophysical consequences. Fowler [1926] applied them to explain the stability of white dwarf stars: the electron degeneracy pressure arising from the Exclusion Principle resists gravitational collapse even at zero temperature. Sommerfeld [1927] resolved the long-standing puzzle of why conduction electrons

contribute far less to metallic heat capacity than classical equipartition predicts: at room temperature, $k_B T \ll E_F$, and the Pauli exclusion principle suppresses thermal excitations for all but the small fraction of electrons within $k_B T$ of the Fermi surface.

9.3 Dirac's Relativistic Derivation

Paul Dirac derived the relativistic quantum equation for the electron in 1928 [Dirac, 1928]. The Dirac equation requires the wave function to be a four-component spinor, automatically producing:

- Intrinsic spin-$\frac{1}{2}$: the spin degree of freedom is not added by hand but emerges from the algebra of the equation.
- $g_s = 2$ exactly: the anomalous gyromagnetic ratio of Uhlenbeck and Goudsmit is a direct consequence of the Dirac equation; quantum electrodynamic corrections produce the small deviation $g_s = 2.00232\ldots$

Dirac's derivation retroactively justified the Uhlenbeck–Goudsmit proposal and gave Pauli's fourth quantum number its derivation from first principles.

9.4 The Spin–Statistics Theorem

Pauli proved in 1940 [Pauli, 1940] that in any relativistic quantum field theory satisfying locality, causality, and positivity of energy, particles with half-integer spin *must* obey the Exclusion Principle (Fermi–Dirac statistics), while particles with integer spin must obey Bose–Einstein statistics. The Exclusion Principle is thus not an independent postulate in quantum field theory: it is a theorem, a consequence of combining quantum mechanics with special relativity in a consistent way. This is the deepest confirmation of Pauli's 1925 heuristic.

Pauli received the Nobel Prize in Physics in 1945 specifically "for the discovery of the Exclusion Principle, also called the Pauli Principle."

10 Pauli as Machine Learner

The vocabulary of modern machine learning—supervised learning, hypothesis class, inductive bias, feature engineering, model selection—maps onto Pauli's scientific reasoning in almost the same way it maps onto Einstein's photoelectric reasoning.

10.1 The Learning Problem

The "dataset" assembled by Pauli is a table of discrete counts (Table 4) and multiplicities (Tables 5, 6). The task is to find a model that predicts all entries in these tables from a minimal set of structural rules. In modern ML parlance, this is a multivariate *classification* or *generative model* problem over a discrete state space.

10.2 The Hypothesis Class and Inductive Bias

Einstein's hypothesis class was the set of affine functions $f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu$, derived from the light-quantum postulate. Pauli's hypothesis class is the set of binary occupation functions on four-integer-labeled states, derived from two postulates:

1. *Four quantum numbers*: the state space is the Cartesian product $\{1, 2, \ldots\} \times \{0, \ldots, n-1\} \times \{-l, \ldots, +l\} \times \{+\frac{1}{2}, -\frac{1}{2}\}$.
2. *Exclusion*: occupation numbers are in $\{0, 1\}$.

Together these postulates restrict the capacity of each subshell to exactly $2(2l+1)$—a prediction with zero adjustable parameters beyond the specification of the state space.

Prior to Pauli, the effective hypothesis class was a much larger family: models with arbitrary occupation rules, non-integer quantum numbers, or additional continuous parameters. Pauli's postulates function as an extreme inductive bias that collapses all combinatorial freedom into a single prediction. In learning-theoretic terms, the VC dimension of Pauli's "model class" (the lattice of binary-occupied states subject to the exclusion constraint) is effectively 1: there is only one model in the class (given the four quantum numbers and the exclusion rule, the capacity function is uniquely determined), so any single observation that confirms the predicted capacity suffices to "learn" the model.

10.3 Feature Selection

Pauli's four quantum numbers are four *features* of each electron's state. By 1924, three of these features were known from Sommerfeld's work; the problem was that the known feature set (n, l, m_l) was incomplete. Pauli's scientific contribution was to recognise that a fourth binary feature is needed—not because a large dataset demanded it, but because the observed capacity data were off by the precise factor of 2 that a binary

feature adds.

This is a form of *feature engineering*: diagnosing that the existing feature set is insufficient, inferring the minimum additional feature—a two-valued one—needed to close the discrepancy, and verifying that the augmented feature set fits all observations exactly.

10.4 What a Naive Data-Driven Learner Would Miss

A purely data-driven learning system confronting Table 4 would observe ten (subshell, count) pairs. A regression tree or neural network could be trained to predict these counts from (n, l). What it could not do without theoretical structure:

1. It would not know that the correct model has the form $2(2l + 1)$—it would fit a table of numbers and perhaps find the pattern numerically.
2. It would not know that the explanation of the pattern is a fourth quantum number with exactly two values. The data are consistent with many mathematical formulas; the fourth quantum number is the physically and theoretically minimal explanation.
3. It would not generalise to the anomalous Zeeman effect predictions without the physical model as a common underlying theory. The connection between shell capacity data and Zeeman splitting data requires the quantum number structure as the shared latent variable.
4. It could not predict the Fermi distribution (13) or white dwarf stability—applications that rely on the Exclusion Principle in a continuum of energies, far outside the discrete atomic-level dataset.

Pauli's irreplaceable contribution was the theoretical unification: a single two-sentence postulate that simultaneously explains the capacity data, the spectral multiplicities, and the Zeeman patterns, and that generalises to all subsequent applications of the Exclusion Principle in condensed matter physics, nuclear physics, and quantum field theory.

10.5 Transfer Learning

Einstein's slope h/e transferred exactly from sodium to lithium—a zero-shot prediction confirmed to 0.2%. Pauli's shell-capacity formula $2n^2$ transfers from atoms to molecules (Hückel's rule for aromatic stability), to nuclei (nuclear shell model), to solid-state crystals (Fermi surfaces), and to neutron stars (neutron degeneracy pressure). Each application is zero-shot: the Exclusion Principle was not re-estimated but transferred with no adjustment, constrained only by the energy levels applicable in the new domain.

11 Comparison with the Photoelectric Episode

Table 7 compares the two episodes side by side.

A key structural difference is the *type of data*. Einstein's equation is a linear regression on continuous data (ν_i, V_{0i}); the free parameters $\beta = h/e$ and $\alpha = -W/e + v_c$ are estimated by OLS/MLE with degrees of freedom from three or more observations. Pauli's model involves a discrete counting argument; the "free parameters" are the integer ranges of n, l, m_l, and the binary range of m_s, and they are identified by qualitative and semi-quantitative spectroscopic data rather than by a quantitative regression.

A second structural difference is the *role of error*. Einstein's regression has a genuine residual ε from measurement noise; Millikan's precision was finite and the standard error of h/e was estimated at 0.1%. Pauli's counting formula predicts integers; there is no room for statistical noise—the capacity of the $2p$ subshell is exactly 6, not 6.1 ± 0.3. The "goodness of fit" is either perfect or it is not.

In both cases, theory supplied the hypothesis class before quantitative data were available: Einstein wrote down $V_0 = (h/e)\nu - W/e$ in 1905 before clean data existed; Pauli wrote down the exclusion rule and the fourth quantum number in January 1925, relying principally on the qualitative patterns in Stoner's (1924) table. In both cases, the theoretical prior reduced the effective number of free parameters to near zero, enabling confirmation with a minimal dataset.

12 Pauli's Epistemic Situation

Pauli's relationship to his own principle shares certain features with Millikan's relationship to Einstein's equation. As Millikan confirmed Einstein's equation while for years publicly rejecting its physical interpretation, Pauli proposed the exclusion principle and the fourth quantum number while explicitly refusing to interpret m_s physically:

> "Already in my original paper I stressed the circumstance that I was unable to give a logical reason for the exclusion principle or to derive it from more general assumptions... To me, the suggestion of Stoner appeared to be an important step forward, since for the first time the number 2, characteristic of the closed shells, had to be attributed to the electron itself—to its 'two-valuedness.'" [Pauli, 1964]

The lesson is the same as in the photoelectric case: a sufficiently parsimonious formal model can unify a body of empirical observations before the physical mechanism underlying the model is understood. Pauli in 1925 was, in modern parlance, fitting a discrete probability model over quantum state space; the physical object that corresponds to m_s—intrinsic spin—was supplied by Uhlenbeck and Goudsmit a few months later, and its relativistic necessity was established by Dirac three years after that.

When Uhlenbeck and Goudsmit proposed electron spin, Pauli was initially sceptical, pointing out that a classically spinning electron the size of a point particle would need a surface velocity exceeding the speed of light. This reservation was eventually resolved by Dirac's equation, in which spin is a purely quantum-mechanical property with no classical analogue—exactly the "classically non-describable two-valuedness" Pauli had insisted upon from the start.

13 Conclusions

Pauli's 1925 determination of the quantum-mechanical state space for atomic electrons is a canonical exercise in scientific model specification. Starting from a handful of empirical regularities—Stoner's factor-of-2 per sublevel, the doublet structure of alkali spectra, the anomalous Zeeman multiplicities, and the noble-gas closure counts—Pauli inferred the existence of a fourth, two-valued quantum number and a binary occupation constraint. His model has the structure of a four-parameter discrete probability distribution over a

lattice, with the unusual property that the parameters are integers (the ranges of n, l, m_l, m_s) and the predictions are exact counts, leaving zero tunable parameters and zero residual degrees of freedom.

Three features of the episode are especially instructive from a statistical standpoint.

First, extreme parsimony. The principle is a single sentence ($n_s \leq 1$), yet it organises the structure of the entire periodic table, explains all atomic spectral multiplicities, predicts the anomalous Zeeman patterns, and implies Fermi–Dirac statistics. The Akaike-information-criterion premium for zero free parameters—relative to any competitor that uses tunable constants—is overwhelmingly in Pauli's favour.

Second, theory-first identification. Like Einstein's photoelectric paper, Pauli's principle was proposed before quantitative experimental confirmation was complete. The fourth quantum number was inferred from the inability of three quantum numbers to explain a well-known discrepancy (factor of 2), not from a new precision measurement. The measurement confirmation—via Zeeman pattern counts, X-ray level counts, and chemical periodicities—served to verify a pre-existing theoretical prediction.

Third, uninterpreted success. Pauli in 1925 did not know what m_s was. In this he resembles Planck in 1900, who introduced the energy quantum $h\nu$ to fit the blackbody spectrum without knowing what a "quantum" physically was, and Einstein in 1905, who proposed the light quantum while acknowledging that it conflicted with the wave theory of light. Each of these model specifications—Planck's, Einstein's, Pauli's—introduced a formal device that organised empirical data parsimoniously, long before a self-consistent physical interpretation was available. The pattern suggests that in fundamental physics, as in econometrics, the ability to specify a parsimonious model is often the critical intellectual step, and the ability to interpret its parameters often follows much later.

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Table 7: Structural comparison of Einstein's photoelectric derivation (1905–1916) and Pauli's exclusion principle derivation (1924–1925) from the perspective of statistical inference and machine learning.

Dimension	Einstein (photoelectric)	Pauli (exclusion principle)
Empirical anomaly	Stopping potential independent of intensity; depends on ν	Shell capacity $2n^2$ not n^2; anomalous Zeeman multiplicities
Key precursor data	Lenard (1902): V_0 vs. ν qualitative	Stoner (1924): 2 electrons per m_l sublevel
Theoretical postulate	Light quantum: $E = h\nu$	Fourth quantum number m_s; exclusion constraint $n_s \leq 1$
Hypothesis class	Affine functions: $V_0 = \alpha + \beta\nu$	Binary occupation on four-dimensional integer lattice
Free parameters	2 (slope $\beta = h/e$; intercept $\alpha = -W/e + v_c$)	0 (all capacities determined by counting)
Prediction	$V_0 = (h/e)\nu - W/e$; universal slope across metals	Capacity$(n, l) = 2(2l + 1)$; universal across all atoms and shells
Fit statistic	OLS: $R^2 = 0.99999$, $\hat{\sigma} = 1.4\,\mathrm{mV}$ (Millikan 1916)	$\chi^2 = 0$: exact fit to all 10 subshell observations
Transfer	Na and Li slopes agree to 0.2%	Formula transfers to molecules, nuclei, stars
Physical interpretation delayed	Light quantum debated until Compton (1923)	m_s uninterpreted until Uhlenbeck–Goudsmit (1925)
Deductive grounding achieved	QED: photoelectric equation exact	Dirac (1928) and Pauli (1940) spin–statistics theorem
Nobel Prize	1921 (Einstein, for photoelectric equation)	1945 (Pauli, for exclusion principle)